

Embedded speed control of BLDC motors using LPC1549 microcontroller

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Abstract - This paper presents a speed controller of a Brushless DC motor using root locus method. The control algorithm of the Brushless DC motor is implemented on LPC1549 microcontroller, using a hardware event driven state machine. This algorithm runs on a state configurable timer peripheral, without any intervention of the processor. The Hall sensors of the low cost motor are software calibrated in order to reduce the output speed noise from the Brushless DC motor. The computational power of the microcontroller is used in order to achieve acceptable performances from a low cost Brushless DC motor.

Keywords—BLDC motor; root locus; software configurable timer; Hall sensor; speed control;

I. INTRODUCTION

The level of automation is increasing rapidly, not only in industrial application, but especially in home automation and smart city application. One of the most important parts of a control loop system is the actuator. Brushless DC motor is becoming increasingly popular in home automation, smart city applications, power tools or HVAC applications, due to the lower maintenance and lower electromagnetic noise than the DC brushed motor [1].

There are two main control methods of the BLDC motors: sensorless and sensor mode [2-5]. On the one hand the sensorless mode has the advantage of a lower overall cost, because no external parts are needed, but on the other hand the sensorless mode can't be used at low speeds from 0 to 50 RPM due to the low amplitude on the current and the back electromotive force signal. Sensorless mode is normally used for high speed control of the BLDC motors, in applications such as: drones and aeromodels. Position control and low speed control of a BLDC motor requires a position sensor attached to the motor such as an encoder, or hall sensors.

The Brushless DC motors are widely used in industrial applications for variable speed drive systems [6-11]. The most common speed controller used in industrial applications is the PI controller. The PI controller is usually tuned using trial and error method, or calculated using Ziegler-Nichols method. This paper presents the root locus method in order to calculate the PI speed controller of the Brushless DC motor.

The speed controller presented in this paper is implemented on LPC1549 microcontroller from NXP. This microcontroller is specially designed for motor control applications. The most important peripheral of the microcontroller is the State Configurable Timer (SCT). This peripheral is designed as a

time-capture unit combined with an event driven state machine [12]. The BLDC motor control based on Hall sensors, can be implemented entirely on the SCT peripheral, without any intervention of the CPU.

The BLDC motor can be chosen to meet the application specifications. However, there is a lot of cheap BLDC motors on the market, which has some manufacturing flaws. One of the most important manufacturing flaws, is the placement of the Hall sensors. Because of the low quality of the BLDC motor the Hall sensors are not perfectly symmetrical at 120 degrees, and they are not perfectly concentric with the rotor shaft. However, this manufacturing flaws can be minimized using a software calibration of the Hall sensor placements. The software calibration was applied to a BLDC motor, 42BLF01 from Wantai Motors, with great results.

The BLDC motor used in this study was approximated with a first order linear system response, and identified using the step response method. The PI speed regulator was calculated using the root locus method.

II. MATHEMATICAL MODEL OF BLDC MOTOR

First step in designing and developing of the BLDC motor control algorithm, is choosing the most suitable mathematical model for the BLDC model. The best way of analyzing the BLDC motor behaviour is the stationary reference frame (abc), where the three phase voltages are forming a 3x1 matrix.

$$\begin{pmatrix} v_a \\ v_b \\ v_c \end{pmatrix} = \begin{pmatrix} R_a & 0 & 0 \\ 0 & R_b & 0 \\ 0 & 0 & R_c \end{pmatrix} \begin{pmatrix} i_a \\ i_b \\ i_c \end{pmatrix} + \begin{pmatrix} L_a & 0 & 0 \\ 0 & L_b & 0 \\ 0 & 0 & L_c \end{pmatrix} \begin{pmatrix} \frac{di_a}{dt} \\ \frac{di_b}{dt} \\ \frac{di_c}{dt} \end{pmatrix} + \begin{pmatrix} K_{e_a} \\ K_{e_b} \\ K_{e_c} \end{pmatrix} \omega_r \quad (1)[13]$$

where v_a , v_b and v_c are the notations for phase voltages, i_a , i_b and i_c , phase currents, R_a , R_b and R_c , the resistance of the each phase of the stator, L_a , L_b and L_c the phase inductance. The back

electromotive force (BEMF) is represented by the third term of the sum, and is the BEMF constant of the each motor: K_{e_a} , K_{e_b} and K_{e_c} , multiplied by the angular speed of the motor: ω_r . The mathematical model of the BLDC motor presented in (1), presumes that the BLDC motor: has a star wound type stator, has a non-salient rotor type and the motor's three phases are perfectly symmetrical [14].

On a BLDC motor only two phases are active a specific time, so the mathematical model presented in the relation (1), can be simplified:

$$v_s = Ri(t) + L \frac{di(t)}{dt} + K_e \omega_r(t) \quad (2)$$

where v_s is the voltage supply to the BLDC motor, $R = \frac{2}{3}(R_a + R_b + R_c)$, $L = \frac{2}{3}(L_a + L_b + L_c)$ and $K_e = \frac{2}{3}(K_{e_a} + K_{e_b} + K_{e_c})$.

The mathematical model for the mechanical part of the BLDC motor is presented in (3). The Coulumb friction is neglected, the torque constant is notated with K_T , and the T_m represents the mechanical torque of the motor [15]. The rotor inertia is notated with J , and B represents the viscous friction of the motor.

$$K_T i(t) = T_m(t) + J \frac{d\omega_r(t)}{dt} + B \omega_r(t) \quad (3)[16]$$

A. Simplified mathematical model of the BLDC motor

The open loop transfer function of the BLDC motor is obtained from relations (2) and (3). The second order transfer function of the BLDC motor is:

$$H_m(s) = \frac{K_{PWM} K_T}{(Ls + R)(Js + B) + K_T K_e} \quad (4)$$

The datasheet of the BLDC motor does not provide all the parameters needed for generating the simplified mathematical model of the BLDC motor, so a practical identification of the motor is needed in order to approximate the model of the actual BLDC motor. The electrical time constant of the BLDC motor is usually close to the mechanical time constant, so the open loop transfer function can be approximated with a first order system:

$$H_m(s) = \frac{K_m}{(T_m s + 1)} \quad (5)$$

B. Identification of the BLDC motor

The open loop of the BLDC motor is approximated with a first order linear system. The step response method is used to identify the motor. The input of the motor, which is the PWM duty cycle, is stepped up from 40% to 60% and the output of the motor, the speed of the motor in RPM is then recorded. The output of a step response is presented in Figure 1.

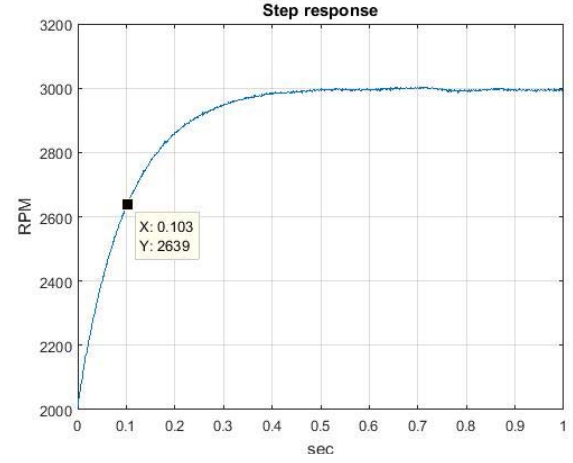


Fig. 1. Identification step response.

The identification of the BLDC motor using the step response method provided the time constant of the motor: $T_m = 103$ ms, and the proportional factor of the motor: $K_m = 18.72$.

III. SPEED CONTROLLER USING ROOT LOCUS METHOD

The speed controller proposed for this study was a regular PI regulator, calculated using the root locus method. In order to use the root locus technique the open loop transfer function of the system and the regulator is needed. The PI regulator was proposed for controlling the speed of the BLDC motor.

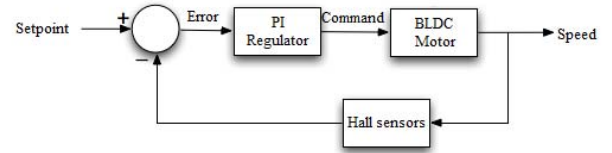


Fig. 2. Block diagram of a PI control algorithm.

The transfer function of the PI regulator, and the open loop transfer function of the motor and the regulator are:

$$H_R(s) = K_p \frac{T_i s + 1}{T_i s} \quad (6)$$

$$H_{ol}(s) = H_R(s) * H_m(s) = \frac{K_m K_p (T_i s + 1)}{T_i s (T_m s + 1)} \quad (7)$$

The open loop transfer function of the motor and the PI regulator has two poles and one zero. The system is stable if the zero is smaller than the smallest pole of the system: $-\frac{1}{T_i} < -\frac{1}{T_m}$, which that $T_i < T_m$. In our case, we choose, $T_i = T_m/20 = 5,2\text{ms}$ in order to make the brushless DC motor much faster. The root locus of the open loop transfer function is presented in Fig 3.

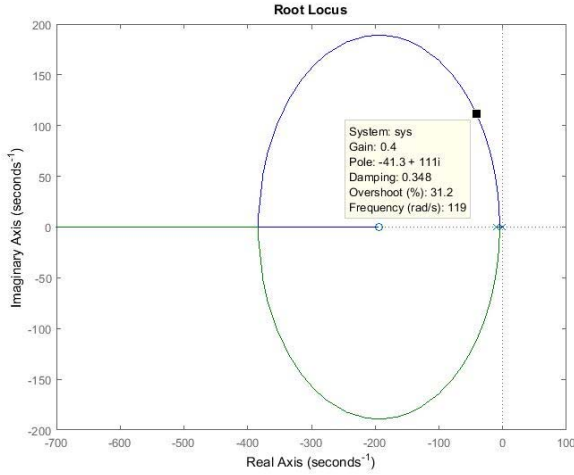


Fig. 3. Root locus of the open loop transfer function.

The proportional gain of the PI regulator is the gain factor from the root locus diagram which offers a robust speed controller, able to work with the BLDC uncertain parameters. In order to find the most robust controller on the root locus diagram, a tangent line to the root locus diagram starting from origin, has to be drawn. The intersection of the root locus with the tangent line is the most robust speed controller for the BLDC motor. In our case: $K_p = 0.4$.

IV. PRACTICAL IMPLEMENTATION

The speed controller was implemented on LPC1549 microcontroller, which is a specialized motor control microcontroller from NXP. The most important feature of the microcontroller is the State Configurable Timer (SCT), which allows the developer to implement complex time based state machine on a single peripheral of the microcontroller without any intervention of the CPU. The BLDC motor control algorithm was implemented on one of the SCT timers using the Hall sensors. Each transition of the Hall sensors is considered as an event for the state machine. There are six possible states corresponding to each possible configuration of the Hall sensors. In each state the two of six outputs (SW1 – SW6) of the electrical driver are active. A detailed state machine diagram of the BLDC motor algorithm is presented in Fig. 4

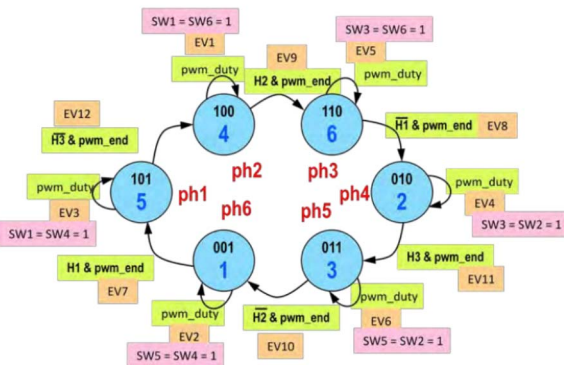


Fig. 4. State machine diagram of the BLDC motor control algorithm [12].

The LPCXpresso Motor Control Kit was used to drive the BLDC motor from an electrical point of view. This platform is a universal electrical driver for low voltages motors based on NXP's microcontrollers. It supports voltages up to 30V, and rated power of the motor, up to 300W. The platform supports motors with up to 4 phases, and provides phase voltage measurement, in-phase current measurement, overcurrent protection and some active break functionality. Also the platform offers encoder and Hall sensor input capability, but it can also be used in sensorless motor control algorithms, using the backelectromotive voltages or the current measurements from each phase of the BLDC motor.

The BLDC motor is a Wantai Motors, 42BLF01 motor with Hall sensors attached to it. This motor has three phases and four pairs of poles, and it offers up to 6000 RPM, with a nominal voltage of 24V. This is a relatively cheap motor, so there have been noticed a slight inaccuracy of Hall sensor placement. Because of this manufacturing flaw, the output speed has lot of noise, which can the BLDC motor in an unstable system. A software calibration of the Hall sensors was successfully implemented. For a full mechanical revolution of the motor, the Hall sensors give 24 transitions: 6 transitions per electrical cycle, times 4 pole pairs of the BLDC motor. After the startup of the motor, the calibration algorithm waits for the motor speed to stabilize, then each hall sensor transition is calibrate according to the average of the last 24 Hall transitions. The calibration values are stored in a lookup table and used at each Hall transition. The software calibration of the Hall sensors, reduced the output speed noise from hundred of RPM from peak to peak, to 10 RPM from peak to peak. The Hall sensor calibration allows the speed of the BLDC motor to be adjusted faster, and more accurately. The time between two transitions of the Hall sensor is measured with the time capture functionality of the SCT timer, without any intervention of the processor, and then converted into RPM, inside the microcontroller.

The actual speed of the microcontroller is transmitted to Matlab program via serial communication. The Matlab data acquisition can be up to 5 kilo samples per second, using a serial communication with a baud rate of 460800 bauds. The data acquisition sampling rate allows an accurate identification of the BLDC motor time constant. The Matlab R2015a program was used for the data acquisition, offline mathematical operations, root locus diagrams and graphs plotting.

V. EXPERIMENTAL RESULTS

The first test of the BLDC speed controller was a no load test. The speed controller was tested using a step up speed reference from 2000RPM to 3000RPM. The digital PI controller was running with a frequency of 1 kHz, and the PI controller coefficients calculated using root locus method were: $K_p = 0.4$ and $T_i = 5.2$ ms. The settling time of the speed controller was 210 ms, while the overshoot was 33.5%. The output speed of the BLDC motor is presented in Fig. 5

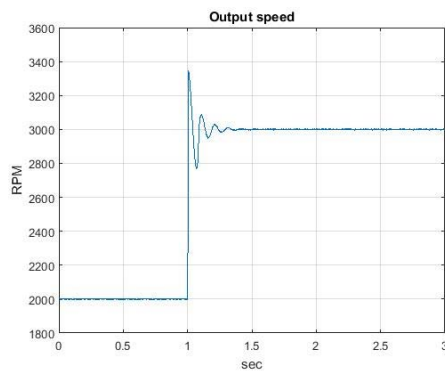


Fig. 5. Idle test output speed

The second test of the speed controller, was under load. The same speed controller with the same parameters was used. The settling time dropped up to 80 ms, but the overshoot increased up to 50.5%. The output speed of the second test is presented in Fig. 6.

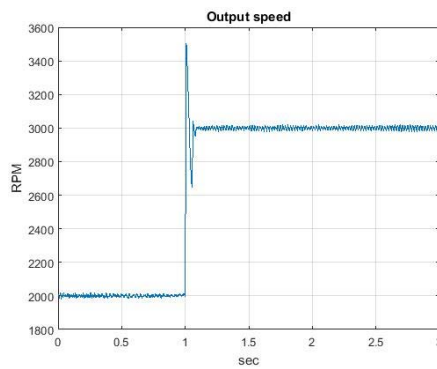


Fig. 6. Load test output speed

The experimental setup included: one DC power supply adjustable up to 30 V and 10 A, the LPCXpresso Motor Control Kit platform, LPCXpresso 1549 board and a laptop with Matlab R2015a program for data acquisition. The BLDC motor was powered with 24V according with the datasheet, and it consumes approximately 0,2 A while idling, and up to 1,5 A under load. The entire setup is presented in Fig. 6.

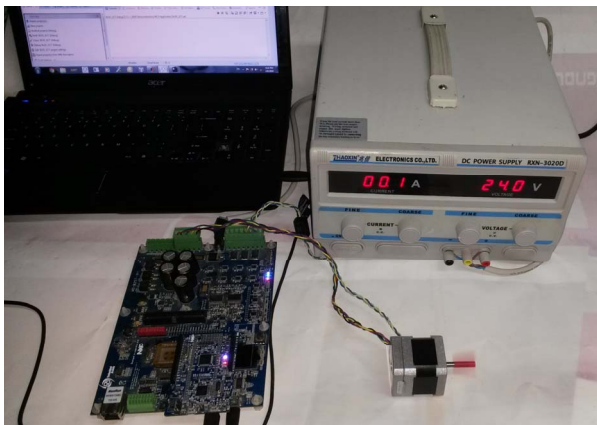


Fig. 7. Experimental setup

VI. CONCLUSIONS

The implemented speed controller offered acceptable results, although the overshoot is a little bit high. The time constant of the Hall sensors is proportional with the motor speed, and the whole system becomes slower. For the future a predictive control of the BLDC motor might reduce the overshoot for the lower speeds of the motor.

Software calibration of the Hall sensors provided a great result, canceling almost the entire speed noise of the BLDC motor. This proves the fact that low priced motors can offer acceptable results using more computational power, which lately is becoming cheaper and cheaper.

REFERENCES

- [1] Dariusz Świerczyński, "Direct Torque Control with Space Vector Modulation (DTC-SVM) of Inverter-Fed Permanent Magnet Synchronous Motor Drive", Ph.D. Thesis, Varsovia, 2005..
- [2] M.L. Corradini, A. Cristofaro, "Fault-tolerant dynamic control allocation for permanent magnet synchronous motors", IEEE 2015 American Control Conference (ACC), Pages: 1585-1590, Published: 2015
- [3] C. Rusu, I. Birou, M.M. Radulescu, A. Bara, "Developing Embedded Control System Platform for Testing PMSM Drives", IEEE 2014 International Conference and Exposition on Electrical and Power Engineering (EPE), Pages: 677-682, Published: 2014
- [4] Antony, T. Cicily, Varghese, M. Beena, Gopinath, Anish, "Poly Phase BLDC Motor Drive with Ten Step Commutation", IEEE 2014 Annual International Conference on Emerging Research Areas: Magnetics, Machines and Drives (AICERA/ICMMD), Article Number: 392, Published: 2014
- [5] U. Karapinar, Z. Esen, M. Şahin, F. Kanburoğlu, "Cascaded Controller Algorithm Design for a Brushless DC Motor with Matlab/Simulink and MCU Implementation", IEEE 2013 21st Signal Processing and Communications Applications Conference (SIU), Book Series: Signal Processing and Communications Applications Conference, Published: 2013
- [6] M.L. Corradini, G. Ippoliti, S. Longhi, G. Orlando, "A Quasi-Sliding Mode Approach for Robust Control and Speed Estimation of PM Synchronous Motors", IEEE Transactions on Industrial Electronics, Volume: 59, Issue: 2, Pages: 1096-1104, Published: FEB 2012
- [7] C. Olivieri, L.D. Leonardo, G. Fabri, M. Tursini, "A DSP-Based Real-Time Simulation Equipment for Fast Motor Control Development", IEEE 2012 5th European DSP Education and Research Conference (EDERC), Pages: 208-212, Published: 2012
- [8] L. Li, Z.M. Peng, B.C. Zhang, Z. Tie, "Rapid Control Prototyping Design of DSP-Based Motion Controller", Advanced Materials Research, Vols. 383-390, pp. 308-314, 2012
- [9] P. Bastiani, M.L. Corradini, G. Ippoliti, S. Longhi, G. Orlando, G. Vannucci, "Experimental Validation of a Sliding Mode Controller for Permanent-Magnet Synchronous Motors" 18th Mediterranean Conference on Control and Automation, Book Series: Mediterranean Conference on Control & Automation, Pages: 527-532, Published: 2010
- [10] D. Mahayana, S. Anwari, "Brushed DC Motor Controller Using Robust Indirect Adaptive Control", IEEE 2009 International Conference on Electrical Engineering and Informatics, VOLS 1 AND 2, Pages: 242-246, Published: 2009
- [11] A. Pisano, A. Davila, L. Fridman, E. Usai, "Cascade control of PM-DC drives via second-order sliding mode technique", IEEE 2008 International Workshop on Variable Structure Systems, Pages: 268-273, Published: 2008
- [12] Application Note "AN11523: BLDC motor control using the LPC15xx STime/PWM", Rev 3.1 – 17 April 2014, © NXP B.V. 2014, http://www.nxp.com/documents/application_note/AN11523.zip
- [13] M. Dobra, I. V. Sita, and P. Dobra, "Efficient implementation of sliding mode control for bldc pm motor using tms320f28335 microcontroller", in

Education and Research Conference (EDERC), 2014 6th European Design in IEEE, 2014 pp 70-74.

- [14] N. Murugananthan, "State Space Modeling and Simulation of Sensorless Permanent Magnet BLDC Motor", International Journal of Engineering Science and Technology, Vol. 2(10), 2010, 5099-5106
- [15] R. Duma, P. Dobra, D. Petreus, I. Sita, and R. Munteanu, "Real-time BLDC motor control using the stellaris lm3s8962 microcontroller," in *Education and Research Conference (EDERC), 2010 4th European*,
- [16] Mirela Dobra, Ioan Valentin Sita, "Fast Evaluation of Advanced Control Structures Based on Rapid Prototyping", Mediamira Science Publisher, Volume 56. Number 1-2. 2015